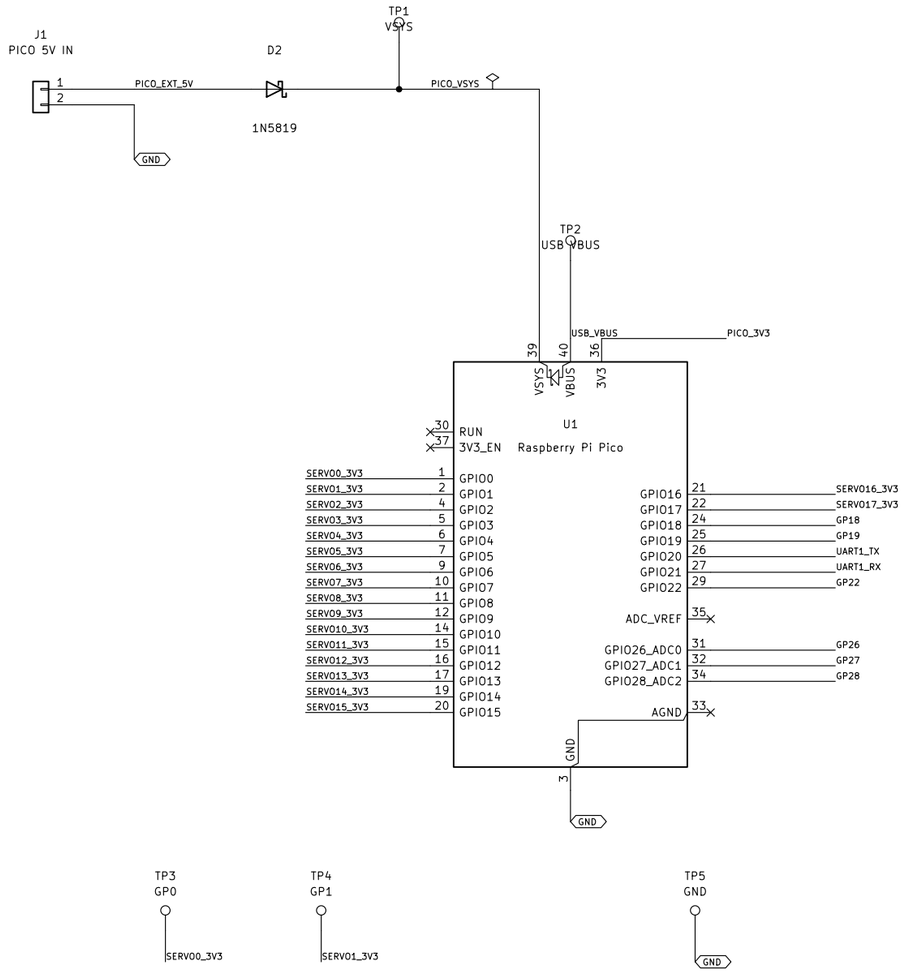
PICO BASEBOARD / 18-CHANNEL MODULAR INTERFACE

Three keyed 2x6 signal ports. Short cables. No motor power on signal connectors.

External Pico power follows Pico datasheet section 4.5, figure 16. USB may remain connected.

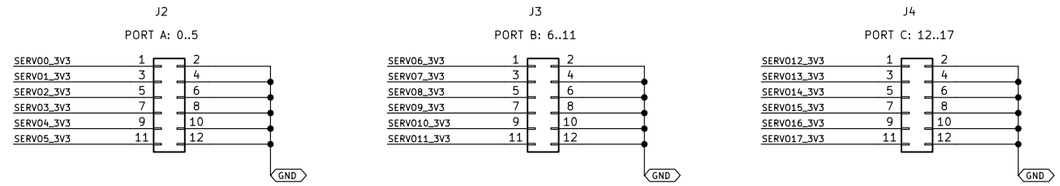


Use two 1x20 female sockets. Pico USB connector at board edge.

No external connection between VBUS and VSYS. D2 cathode band faces VSYS.

Pico pin 3 represents internally common GND pins. AGND is not externally routed.

Reserved outputs and peripherals are routed, not enabled by the current firmware.

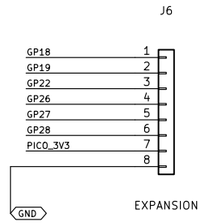


Odd pins 1/3/5/7/9/11 = S0/S1/S2/S3/S4/S5; all even pins = GND.



1 TX -> adapter RX; 2 RX <- adapter TX; 3 GND.

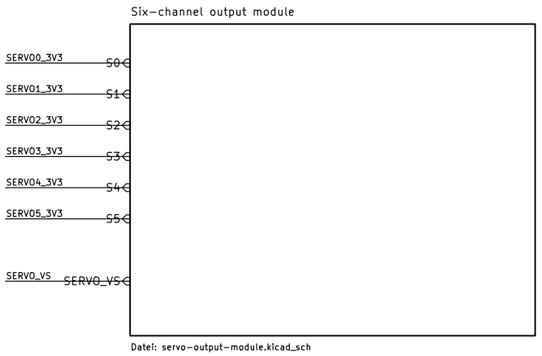
Present firmware uses TX only. Adapter in 3.3V TTL mode.

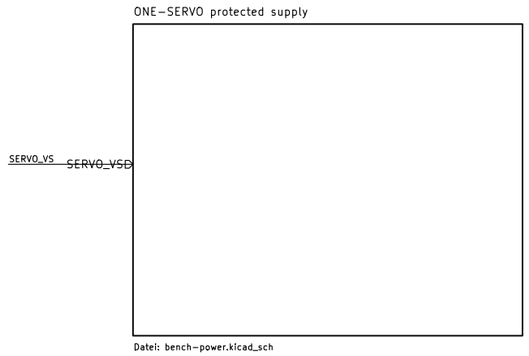


Sheets 2/3 show the first module on Port A and the ONE-SERVO bench supply.

Replicate the module for Ports B/C after current, wiring and clamp sizing are verified.

The hierarchical links model external cables. These are separate physical assemblies.





Modular bench system; unbuilt; no routed PCB

HexaRoam

Sheet: /

File: pico-servo-system.kicad_sch

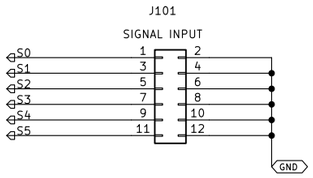
Title: Pico baseboard - 18 servo signals

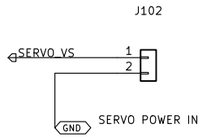
Size: A3	Date: 2026-10-27	**Rev: C - PROPOSAL**
KiCad E.D.A. 10.0.7		Id: 1/3

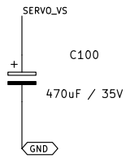
SIX-CHANNEL OUTPUT MODULE / POPULATE CHANNEL 0 FIRST

Reuse on Port A, B or C. S0..S5 are local channel numbers, not hardwired Pico GPIOs.

Only one servo may be powered from the present bench supply. Channels 1..5: DNP.







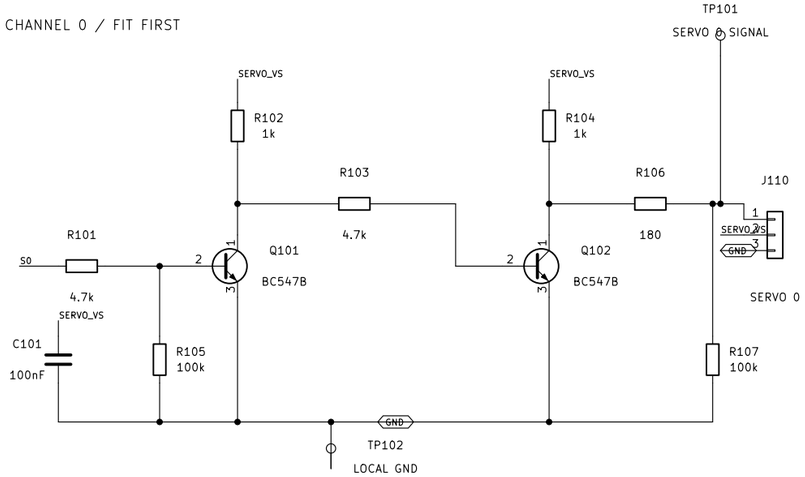
Input: odd pins = S0..S5; even pins = GND.

Each servo: pin 1 SIGNAL, pin 2 +VS, pin 3 GND.

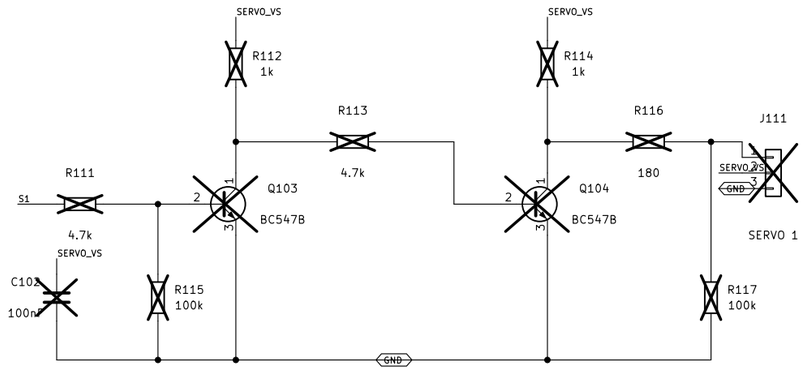
Reinforced power bus and connector ratings: TBD for six populated channels.

Current transistor buffer is intended for the -5V bench rail; re-evaluate for higher servo supplies.

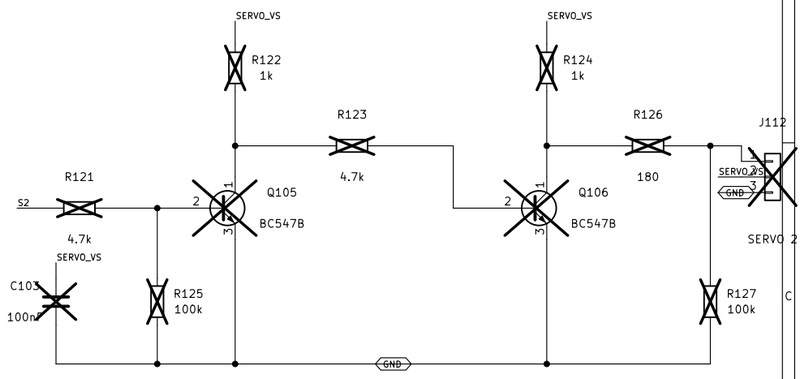
CHANNEL 0 / FIT FIRST



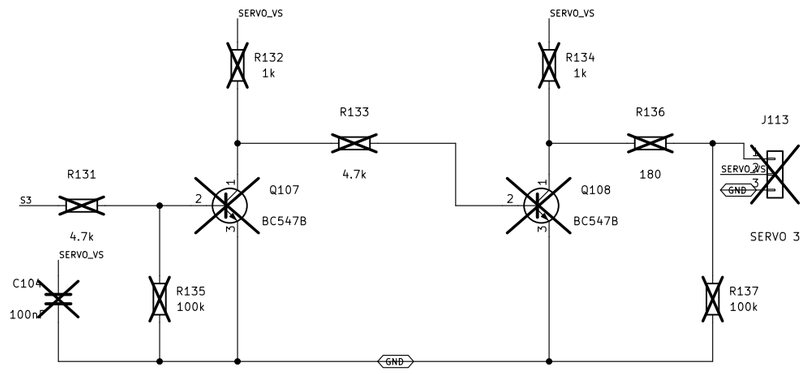
CHANNEL 1 / DNP - RESERVED



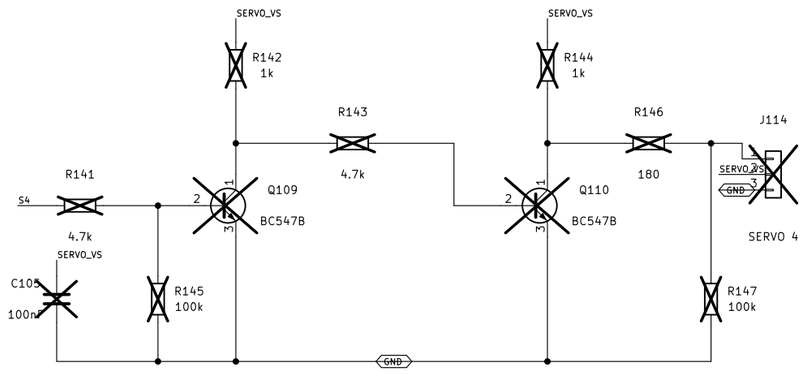
CHANNEL 2 / DNP - RESERVED



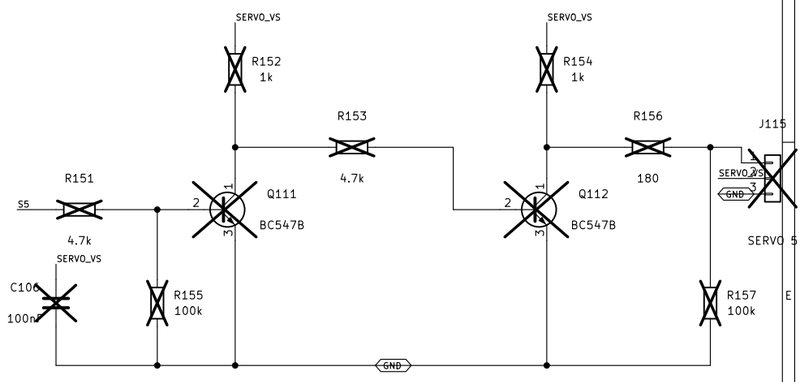
CHANNEL 3 / DNP - RESERVED



CHANNEL 4 / DNP - RESERVED



CHANNEL 5 / DNP - RESERVED



Five crossed-out channels reserve the circuit and board envelope; they are not part of the initial powered build.

No automatic claim of six-servo power capability. Fit only one servo until the power design is sized and tested.

Modular bench system; unbuilt; no routed PCB

HexaRoam

Sheet: /Six-channel output module/
File: servo-output-module.kicad_sch

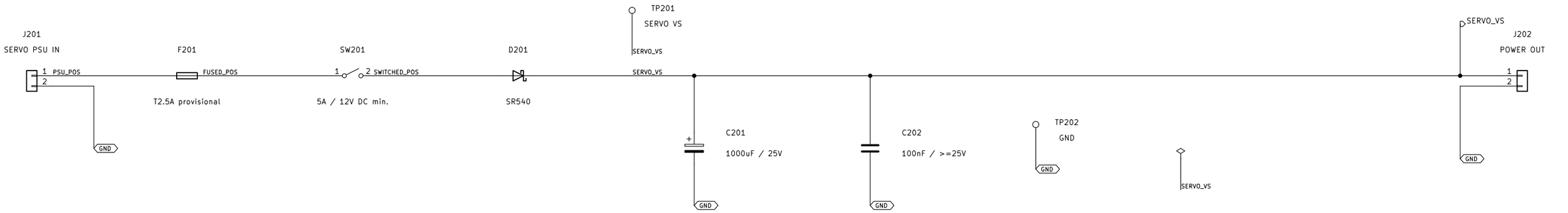
Title: Six-channel signal output module

Size: A3	Date: 2026-10-27	**Rev: C - PROPOSAL**
KiCad E.D.A. 10.0.7		Id: 2/3

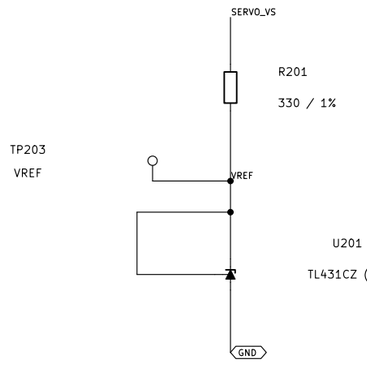
ONE-SERVO BENCH SUPPLY / DISCRETE HYSTERETIC ENERGY DUMP

UNBUILT PROPOSAL: verify startup, clamp response and heating before connecting a servo.

Nominal ~6.03V ON / ~5.76V OFF. Actual limits depend on tolerances and switching dynamics.

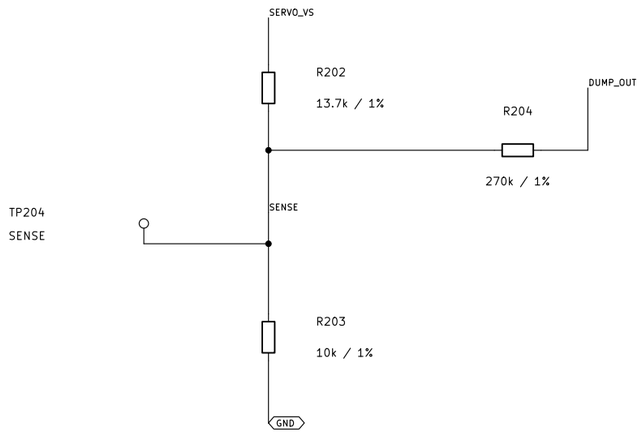


REFERENCE / ~2.495V

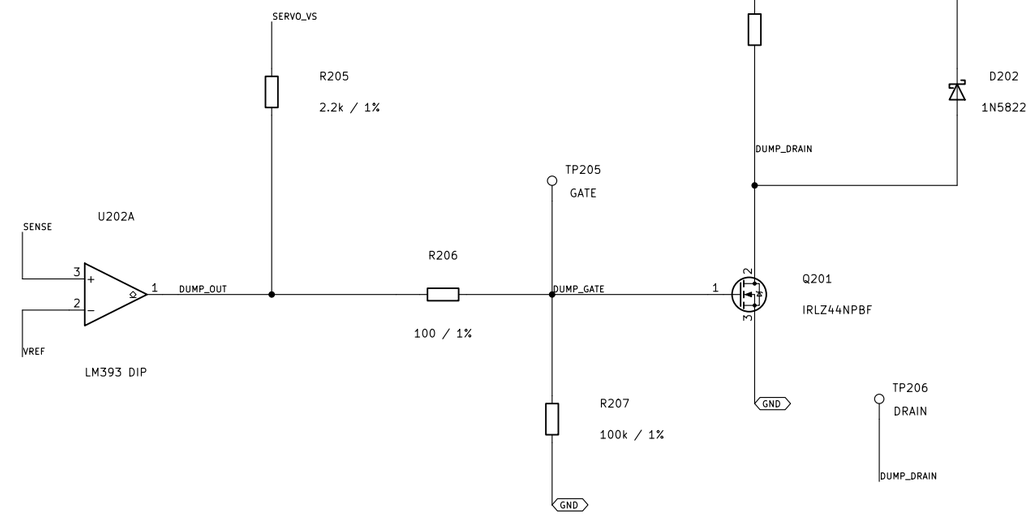


No capacitor directly across TL431.

SENSE / POSITIVE FEEDBACK



COMPARATOR / GATE DRIVE



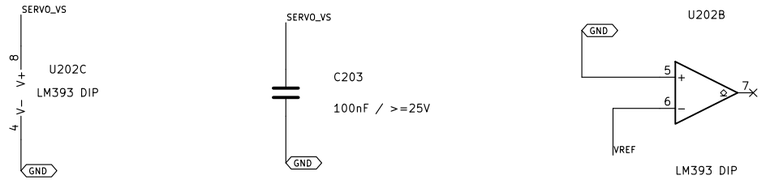
R208: mount OFF BOARD on a suitable heatsink.

6V: ~2.73A / 16.4W. Cooling/temperature qualification pending.

MOSFET tab = drain. Keep dump-current return out of sensing ground.

Both GND paths join locally at capacitor negative / Q201 source.

U202 SUPPLY / UNUSED COMPARATOR



PSU ~5.5V before D201; measure ~5.0-5.3V at the servo under load.

Current-limited startup can produce load conduction or a stalled ramp; measure without a servo.

Not a guaranteed 6V clamp or a six-/eighteen-servo power distributor. See discrete-shunt-design.md.

Modular bench system; unbuilt; no routed PCB

HexaRoam

Sheet: /ONE-SERVO protected supply/

File: bench-power.kicad_sch

Title: ONE-servo supply - discrete energy dump

Size: A3	Date: 2026-10-27	**Rev: C - PROPOSAL**
KiCad E.D.A. 10.0.7		Id: 3/3